

CARLOS MARTÍNEZ

Software Engineer

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ENGINES

Frostbite Unreal Engine Unity

PROGRAMMING

C++ C# C Java Lua

WORKFLOWS

Git Perforce Jenkins Jira Agile Maya

LANGUAGES

Spanish: **Native**
English: **IGCSE English as First Language / IELTS C1**

COURSES

-  **Course in User Experience (UX)**
01/2026, by Escuela de Empresa
-  **Unreal Engine 5**
02/2024, by Carlos Coronado (Udemy)
-  **Unreal Engine 4**
12/2023, by Carlos Coronado (Udemy)
-  **C++ Programming for Unreal Game Development Specialization**
11/2023, by the University of Colorado System (Coursera)
-  **Design, organization and evaluation of video games and gamification**
2017, by the European University of Madrid (Miriadax)
-  **Unity3D**
2016, by the University of León

ABOUT ME

Software Engineer with two MScs, professional experience in AAA game development and a strong foundation in Computer Science and Mathematics. Specializing in UI, gameplay systems, and AI, with a design-oriented mindset.

EXPERIENCE

- C++ Generalist Software Engineer Intern | Electronic Arts**
 07/2025 – Present  Madrid, Spain
 - Working on AAA unannounced games made with **Frostbite and Unreal Engine 5**.
 - Developed modular, data-driven customization features, implementing robust data models, validation policies, and Lua scripting.
 - Architected UI screens using **MVVM, CommonUI and TDD**; refactored and optimized the State Machine and implemented async backend tracking.
 - Engineered menu camera systems and optimized server-client queries. Resolved critical bugs including memory leaks, networking errors, and build-tool crashes.
- Junior Programmer | Maniac Panda Games**
 05/2022 – 09/2023  Remote
 - Worked on **Racerloop**  , an arcade racing game built with Unity.
 - Developed modular gameplay systems across the project, focusing on AI navigation and behavior, synchronous multiplayer networking, automated testing, and math utility libraries.

- Software Engineering Intern at HP SCDS**
 09/2018 – 06/2020  León, Spain
 - Developed **MotionTracker**  , a Python target tracking system.
 - Finalist of the XV "Programa Observatorio Tecnológico HP" awards.** 

EDUCATION

- Master's Degree in Video Game Development | Complutense University of Madrid**
 10/2024 – 07/2025  Madrid, Spain
 - Specialized in **Unreal Engine and C++**, including courses such as Graphics Programming, Artificial Intelligence, Production, Networking, Engine, Physics, UI, and more. **Grade: 10**
- Master's Degree in Video Game Design and Programming | Open University of Catalonia**
 09/2020 – 01/2022  Online
 - Specialized in **Game Design and Unity 2D and 3D programming using C#**. **Grade: 9.55**
- Student Exchange | California State University Monterey Bay**
 08/2019 – 01/2020  California, USA
 - The Art of Storytelling (A+), Graphics Programming (A+), Environment Modeling (A+), Character Design (A+) and Game Design II (A).
- Degree in Computer Science | University of León**
 09/2016 – 06/2020  León, Spain
- Additional Education**
 - Joint studies of **Mathematics and Computer Science** degrees. 93 credits.
 - Baccalaureate in Science and Technology. **With honors**.
 - English-Spanish Bilingual** Primary and Secondary Education.

ACHIEVEMENTS SAMPLE PROJECTS

★ **Stimulation of Mathematical Talent Project (ESTALMAT CyL)**
2009 to 2013

★ **Musician**
Professional Degree in Music, specialty French Horn

★ **Basketball player and coach**
Played the National Championship and won leagues

Check my full portfolio here: [Portfolio](#) 

 **Master's Thesis: Dreadbone, Astral Maw** | [Web](#) 

 12/2024 – 07/2025

- A real-time strategy puzzle game made with **Unreal Engine 5** and **C++**.
- Game Architecture and Planning, AI, UI, Gameplay Systems, Debug Tools, Grid Generation, and other areas.

 **Racerloop, Maniac Panda Games** | [Web](#)  | [Trailer](#) 

 05/2022 – 09/2023

- AI Players, Synchronous Online Game Mode, PlayFab implementation, several Gameplay Systems, Testing Automation, UI screens, Math Utils, Offline Build, Simple VFX...

 **Breathdown** | [Web](#)  | [Itch.io](#)  | [Repository](#) 

 02/2025

- Developed in two days for the *Global Game Jam 2025*.
- Short narrative game. Programmer and assistant designer.

 **The Abyss** | [Web](#)  | [Itch.io](#)  | [Repository](#) 

 11/2024

- Developed in two weeks for *The Confounding Calendar 2022*.
- Short 2D puzzle game. Programmer and designer.

 **Flash your Fears** | [Web](#)  | [Itch.io](#)  | [Repository](#) 

 02/2022

- Developed in one week for *Brackeys' Game Jam 2022.1*.
- Survival 2D game. In charge of overall direction and lead programmer.

 **Master's Thesis: Katergaris' Labyrinth** | [Web](#)  | [Report](#) 

 09/2021 – 01/2022

- 2D *roguelite* dungeon crawler programmed and designed on my own as Master's thesis.
- Procedural generation, stealth, sound, shops, storage and dialogue systems, original narrative, and modular creation of objects, abilities, and armors.

 **MotionTracker, HP SCDS** | [Web](#)  | [Project Proposal](#)  | [HP SCDS Awards](#) 

 09/2018 – 06/2020

- Modular Python-based application capable of tracking moving objects with a camera, even when the camera itself is in motion. Developed iteratively using agile methodologies.
- HP SCDS Internship: Grade of 9.5/10.
- Final Degree Project: Grade of 9.8/10.
- Industry Recognition: Finalist in the HP SCDS Awards.